

Noor Mk 1

MVP Product Requirements

Purpose: Counter-UAS intercept munition for forward/enemy territory engagement

Challenge: Dutch MoD Counter Strike Drone Challenge

Goal: Detect, identify, and disable drones before they reach allied territory

Core Concept

Man-portable autonomous guided munition that intercepts hostile UAS in forward areas. Launches from lightweight tube, acquires target autonomously, and neutralizes threat before it reaches friendly lines.

Minimum Requirements

Form Factor

- Length: ~70cm
- Weight: <15kg (man-portable)
- Launcher: Lightweight tube, single-operator deploy

Detection & Acquisition

- Onboard sensor: Daylight camera + optional IR
- Target acquisition: Autonomous lock after launch
- Range: Acquire target at 1000m minimum
- Target class: Class 1-3 UAS

Guidance & Navigation

- Primary: Visual tracking (onboard camera)
- Backup: INS for GPS-denied environments
- Terminal guidance: Proportional navigation or pursuit
- Update rate: >20Hz guidance loop

Propulsion

- Type: Solid rocket motor
- Speed: >100 m/s (faster than target UAS)
- Endurance: 3-5minute flight time
- Range: 3-3.5km effective

Engagement

- Kill mechanism: Kinetic impact or proximity fragmentation
- Autonomous engagement: No operator input required post-launch
- Time to intercept: <30 seconds at 1km

Communications

- Pre-launch: Target designation from external sensor (optional)
- Post-launch: Fire-and-forget (no guidance link required)
- Telemetry: Optional downlink for BDA

Integration

- Accept target cue from Mukagen or external radar/EO system
 - Standalone capable (self-acquire without external cue)
 - Compatible with standard C2 (ATAK, etc.)
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Key Differentiators (vs. existing systems)

1. **Forward employment:** Engages in enemy territory, not defensive
 2. **Autonomous acquisition:** No operator-in-loop post-launch
 3. **Man-portable:** Single operator launch capability
 4. **GPS-denied capable:** INS fallback for contested environments
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Out of Scope (MVP)

- Swarm coordination (future variant)
 - Multi-target engagement per munition
 - Reusability
 - Subsurface capability
 - Network-enabled guidance updates
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Success Criteria

- ☐ Autonomous target acquisition in flight
- ☐ Successful intercept of Class 2 UAS at 1km
- ☐ Launch-to-kill <30 seconds
- ☐ Operates without GPS
- ☐ Single operator deployment